

Laboratory Work 3

Student: Orynbassar Nurali

Task 1. Generate and insert 200 random rows

I generated and inserted 200 random rows into the airport database.

The screenshot shows the pgAdmin 4 interface. The top menu bar includes File, Object, Tools, Edit, View, Window, and Help. The left sidebar shows the Object Explorer with a tree view containing Servers, lab2, and phonebook. The main pane displays a SQL query in the Query editor:

```
1 SELECT COUNT(*) FROM airport;
```

Below the query editor, the Data Output tab is active, showing the result of the query. The result is a single row with the following data:

	count bigint
1	205

The bottom status bar indicates "Showing rows: 1 to 1" and "Page No: 1 of 1".

Task 2. Add KazAir

I inserted a new airline called KazAir with Kazakhstan as the country

The screenshot displays the pgAdmin 4 interface. On the left, the Object Explorer shows a tree structure with 'Servers', 'lab2', and 'phonebook'. The main pane shows a SQL query being executed in the 'Query' tab. The query is an INSERT statement into the 'airline' table, with values for 'airline_id' (selected as MAX(airline_id) + 1), 'airline_code' ('KZ'), 'airline_name' ('KazAir'), and 'airline_country' ('Kazakhstan'). The 'created_at' and 'updated_at' fields are set to 'CURRENT_TIMESTAMP'. The 'Messages' tab at the bottom shows the execution result: 'INSERT 0 1' and 'Query returned successfully in 59 msec.'. A green notification box at the bottom right confirms the successful execution. The Windows taskbar at the bottom shows the system clock as 20:44 on 28.09.2026.

```
1 INSERT INTO airline
2   (airline_id, airline_code, airline_name, airline_country, created_at, updated_at)
3   VALUES
4   (
5     (SELECT MAX(airline_id) + 1 FROM airline),
6     'KZ',
7     'KazAir',
8     'Kazakhstan',
9     CURRENT_TIMESTAMP,
10    CURRENT_TIMESTAMP
11  );
```

INSERT 0 1
Query returned successfully in 59 msec.

✓ Query returned successfully in 59 msec. ✕

Total rows: Query complete 00:00:00.059

CRLF Ln 11, Col 7

18°C Mostly cloudy 20:44 28.09.2026

Task 3. Update KazAir country

I updated the country of KazAir from Kazakhstan to Turkey

The screenshot displays the pgAdmin 4 web interface. On the left, the 'Object Explorer' shows a tree structure with 'Servers' expanded, containing 'lab2' and 'phonebook'. The main workspace is titled 'airport_db/postgres@lab2'. It features a toolbar with icons for query execution and a 'Query' tab. The query editor contains the following SQL statement:

```
1 UPDATE airline
2 SET airline_country = 'Turkey',
3   updated_at = CURRENT_TIMESTAMP
4 WHERE airline_name = 'KazAir';
```

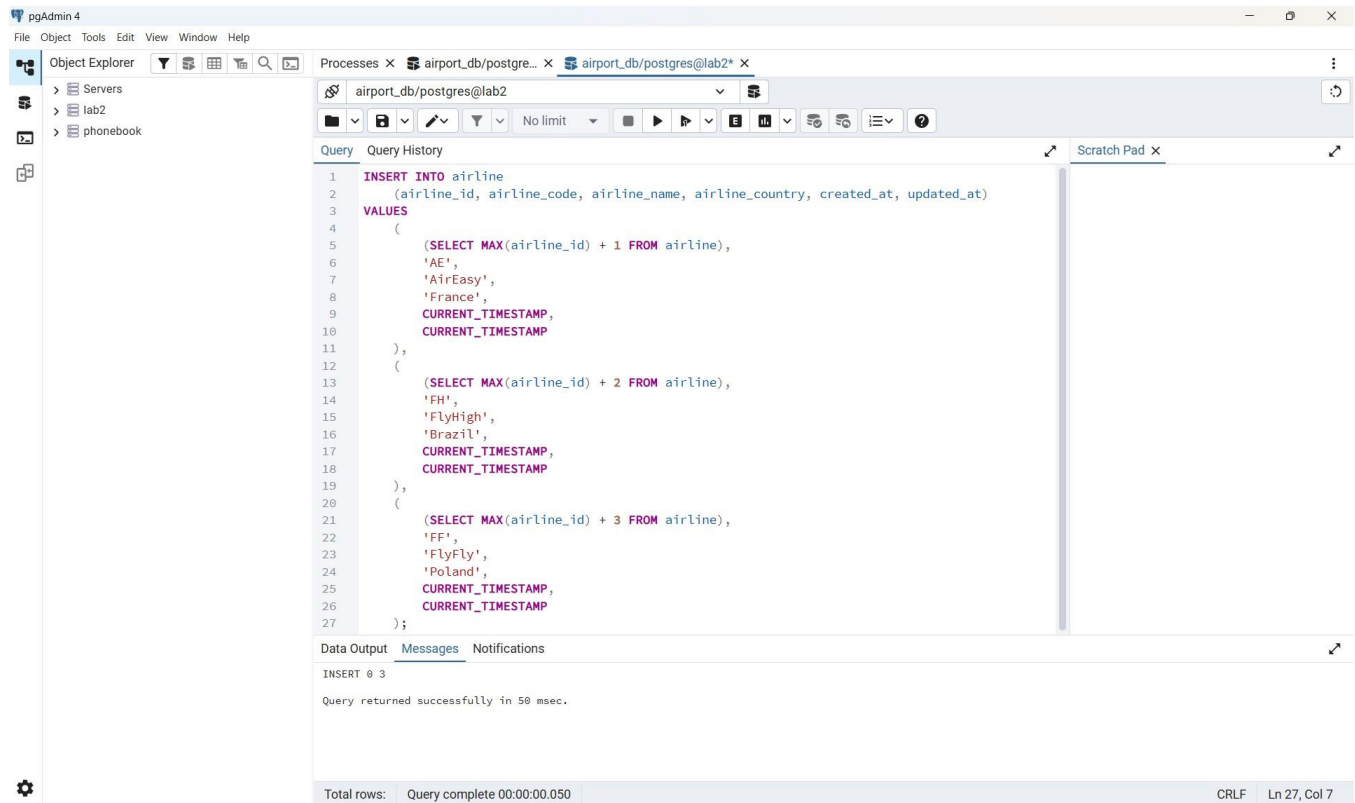
Below the query editor, the 'Messages' tab is active, showing the execution result:

```
UPDATE 1
Query returned successfully in 49 msec.
```

The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.049', 'CRLF', and 'Ln 4, Col 31'.

Task 4. Insert three airlines

I inserted three airlines at once: AirEasy, FlyHigh and FlyFly, with their specified countries.



The screenshot shows the pgAdmin 4 interface with the following components:

- Object Explorer:** Displays a tree view with 'Servers', 'lab2', and 'phonebook'.
- Query Editor:** Contains an SQL query to insert three airlines into the 'airline' table. The query uses a multi-row insert with subqueries to generate unique IDs.
- Messages Panel:** Shows the execution result: 'INSERT 0 3' and 'Query returned successfully in 50 msec.'
- Status Bar:** Displays 'Total rows: Query complete 00:00:00.050', 'CRLF', and 'Ln 27, Col 7'.

```
1  INSERT INTO airline
2  (airline_id, airline_code, airline_name, airline_country, created_at, updated_at)
3  VALUES
4  (
5      (SELECT MAX(airline_id) + 1 FROM airline),
6      'AE',
7      'AirEasy',
8      'France',
9      CURRENT_TIMESTAMP,
10     CURRENT_TIMESTAMP
11  ),
12  (
13      (SELECT MAX(airline_id) + 2 FROM airline),
14      'FH',
15      'FlyHigh',
16      'Brazil',
17      CURRENT_TIMESTAMP,
18      CURRENT_TIMESTAMP
19  ),
20  (
21      (SELECT MAX(airline_id) + 3 FROM airline),
22      'FF',
23      'FlyFly',
24      'Poland',
25      CURRENT_TIMESTAMP,
26      CURRENT_TIMESTAMP
27  );
```

INSERT 0 3
Query returned successfully in 50 msec.

Total rows: Query complete 00:00:00.050 CRLF Ln 27, Col 7

Task 5. Delete flights arriving in 2024

I deleted flights whose scheduled arrival time was in 2024. I used a date range from January 1, 2024 to January 1, 2025.

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the Object Explorer with a tree view containing 'Servers', 'lab2', and 'phonebook'. The main window is titled 'airport_db/postgres@lab2' and shows a SQL query editor with the following code:

```
1 DELETE FROM flights
2 WHERE sch_arrival_time >= '2024-01-01'
3 AND sch_arrival_time < '2025-01-01';
```

Below the query editor, the 'Messages' tab is active, displaying the following output:

```
DELETE 0
Query returned successfully in 72 msec.
```

The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.072' and 'CRLF Ln 3, Col 39'.

Task 6. Increase ticket prices by 15%

I increased all ticket prices by 15 percent by multiplying ticket_price by 1.15, and updated the timestamp.

The screenshot displays the pgAdmin 4 interface. On the left, the Object Explorer shows a tree structure with 'Servers', 'lab2', and 'phonebook'. The main pane is titled 'airport_db/postgres@lab2' and contains a SQL query editor with the following code:

```
1 UPDATE booking
2 SET ticket_price = ticket_price * 1.15,
3    updated_at = CURRENT_TIMESTAMP;
```

Below the query editor, the 'Messages' tab is active, showing the execution results:

```
UPDATE 5
Query returned successfully in 52 msec.
```

A green status bar at the bottom right indicates: '✓ Query returned successfully in 52 msec. X'. The bottom status bar shows 'Total rows: Query complete 00:00:00.052' and 'CRLF Ln 3, Col 36'.

Task 7. Delete tickets below 10000

I selected the bookings with ticket_price below 10000 and deleted those records.

pgAdmin 4

FileObjectToolsEditViewWindowHelp

Object Explorer

Serverslab2phonebook

airport_db/postgres@lab2

No limit

QueryQuery History

1SELECT *
2FROM booking
3WHERE ticket_price < 10000;

Scratch Pad

Data OutputMessagesNotifications

Showing rows: 1 to 5 of 1

	booking_id [PK] integer	flight_id integer	passenger_id integer	booking_platform character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone	status character varying (50)	ticket_price numeric (7,2)
1	1	1	1	Website	2026-09-22 07:04:12.790704	2026-09-28 20:50:01.786866	Confirmed	172.50
2	2	2	2	Mobile App	2026-09-22 07:04:12.790704	2026-09-28 20:50:01.786866	Confirmed	322.00
3	3	3	3	Website	2026-09-22 07:04:12.790704	2026-09-28 20:50:01.786866	Confirmed	517.50
4	4	4	4	Travel Agency	2026-09-22 07:04:12.790704	2026-09-28 20:50:01.786866	Pending	598.00
5	5	5	5	Mobile App	2026-09-22 07:04:12.790704	2026-09-28 20:50:01.786866	Confirmed	218.50

✓ Successfully run. Total query runtime: 140 msec. 5 rows affected. ✕

Total rows: 5Query complete 00:00:00.140CRLF Ln 3, Col 28

Task 8. Set missing airline codes to UNK

I replaced NULL airline codes with the default code UNK and updated the timestamp.

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the 'Object Explorer' with a tree view containing 'Servers', 'lab2', and 'phonebook'. The main window is titled 'airport_db/postgres@lab2' and shows a SQL query editor with the following code:

```
1 UPDATE airline
2 SET airline_code = 'UNK',
3     updated_at = CURRENT_TIMESTAMP
4 WHERE airline_code IS NULL;
```

Below the query editor, the 'Messages' tab is active, displaying the following output:

```
UPDATE 0
Query returned successfully in 47 msec.
```

A green status bar at the bottom right indicates: '✓ Query returned successfully in 47 msec. ✕'. The bottom status bar shows 'Total rows: Query complete 00:00:00.047' and 'CRLF Ln 4, Col 28'.

Task 9. Delete old baggage checks

I deleted baggage check records created before June 1, 2023 when their result was still Not checked.

The screenshot displays the pgAdmin 4 web interface. On the left, the 'Object Explorer' shows a tree structure with 'Servers', 'lab2', and 'phonebook'. The main pane is titled 'airport_db/postgres@lab2*' and contains a SQL query editor with the following text:

```
1 DELETE FROM baggage_check
2 WHERE created_at < '2023-01-01';
```

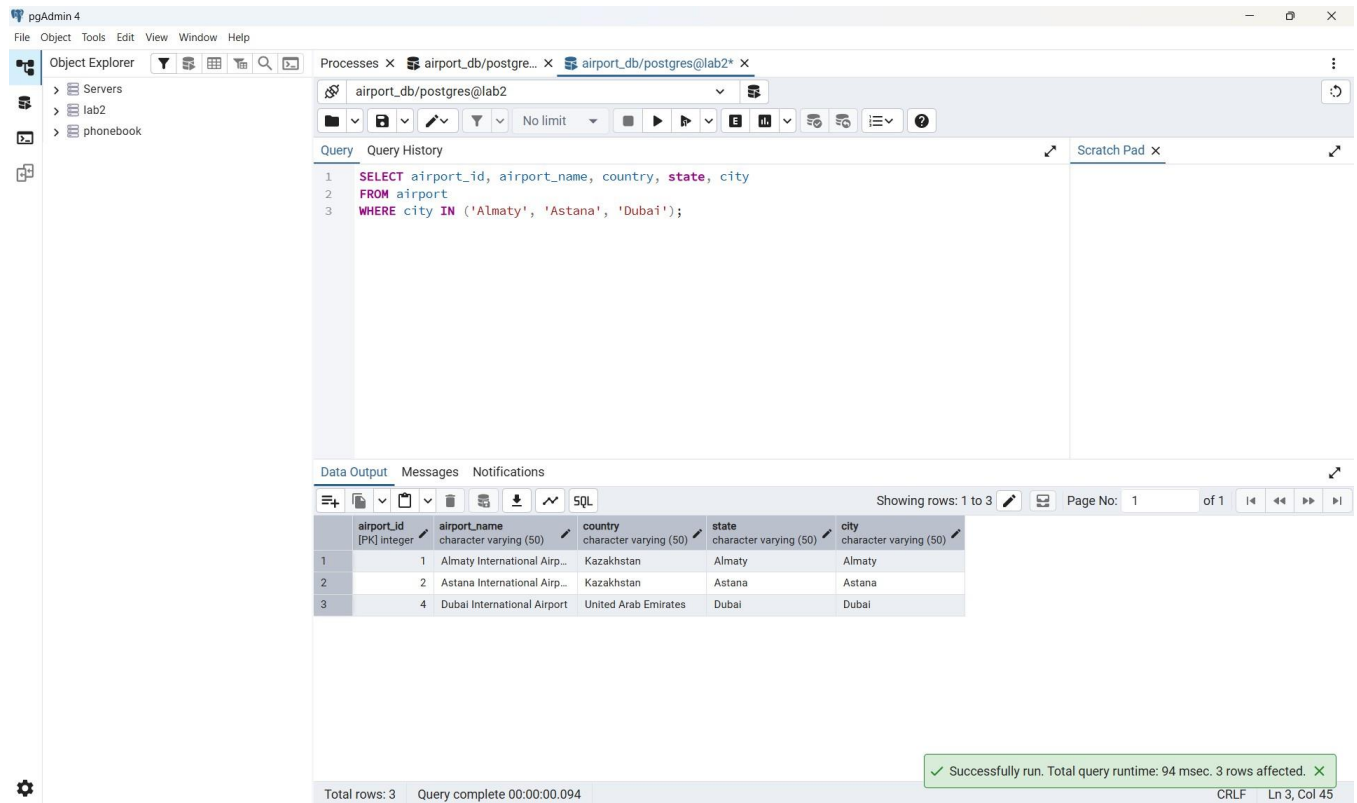
Below the query editor, the 'Messages' tab is active, showing the execution results:

```
DELETE 0
Query returned successfully in 56 msec.
```

A green notification box at the bottom right of the interface states: '✓ Query returned successfully in 56 msec. ✕'. The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.056' and 'CRLF Ln 2, Col 33'.

Task 10. Delete airports with missing state

I removed airport records where state was NULL and the city was one of the specified cities.



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the Object Explorer with a tree view containing 'Servers', 'lab2', and 'phonebook'. The main window is titled 'airport_db/postgres@lab2' and shows a SQL query in the 'Query' tab. The query is:

```
1 SELECT airport_id, airport_name, country, state, city
2 FROM airport
3 WHERE city IN ('Almaty', 'Astana', 'Dubai');
```

Below the query editor, the 'Data Output' tab shows the results of the query. The results are displayed in a table with 5 columns: 'airport_id', 'airport_name', 'country', 'state', and 'city'. The table contains 3 rows of data:

	airport_id	airport_name	country	state	city
1	1	Almaty International Airp...	Kazakhstan	Almaty	Almaty
2	2	Astana International Airp...	Kazakhstan	Astana	Astana
3	4	Dubai International Airport	United Arab Emirates	Dubai	Dubai

At the bottom of the interface, a status bar indicates 'Total rows: 3' and 'Query complete 00:00:00.094'. A green notification box at the bottom right states: '✓ Successfully run. Total query runtime: 94 msec. 3 rows affected. ✕'.

Task 11. Insert baggage inspection and RETURNING

I inserted a new Not checked baggage inspection and used RETURNING to immediately get baggage_check_id and created_at.

The screenshot displays the pgAdmin 4 interface. The left sidebar shows the Object Explorer with a tree view containing 'Servers', 'lab2', and 'phonebook'. The main pane shows the 'Query' tab with the following SQL query:

```
1 INSERT INTO baggage_check
2 (baggage_check_id, check_result, created_at, updated_at, booking_id, passenger_id)
3 VALUES
4 (
5     (SELECT MAX(baggage_check_id) + 1 FROM baggage_check),
6     'Not checked',
7     CURRENT_TIMESTAMP,
8     CURRENT_TIMESTAMP,
9     1,
10    1
11 )
12 RETURNING baggage_check_id, created_at;
```

Below the query editor, the 'Data Output' tab shows the results of the query execution. It displays a table with two columns: 'baggage_check_id' (PK integer) and 'created_at' (timestamp without time zone). The table contains one row with the values 6 and 2026-09-28 20:59:30.496353.

baggage_check_id [PK] integer	created_at timestamp without time zone
6	2026-09-28 20:59:30.496353

At the bottom of the interface, a status bar indicates 'Total rows: 1' and 'Query complete 00:00:00.083'. A green message box on the right states 'Successfully run. Total query runtime: 83 msec. 1 rows affected.'

Task 12. Uppercase airline countries

I converted all airline country names to uppercase using the UPPER function.

The screenshot displays the pgAdmin 4 interface. The left sidebar shows the 'Object Explorer' with a tree view containing 'Servers', 'lab2', and 'phonebook'. The main window is titled 'airport_db/postgres@lab2*' and shows a SQL query editor with the following code:

```
1 UPDATE airline
2 SET airline_country = UPPER(airline_country);
```

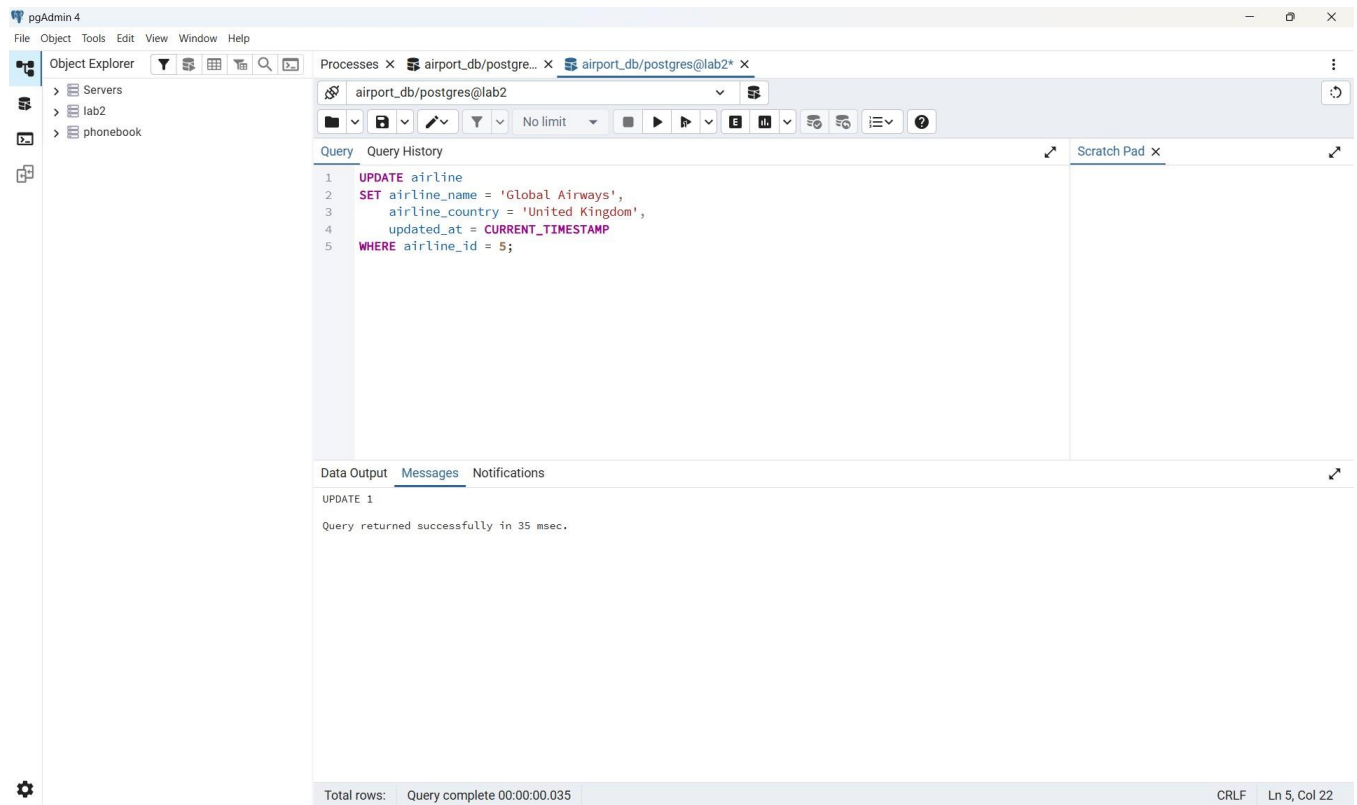
Below the query editor, the 'Messages' tab is active, showing the execution result:

```
UPDATE 9
Query returned successfully in 57 msec.
```

A green notification box at the bottom right of the Messages tab states: '✓ Query returned successfully in 57 msec. ✕'. The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.057' and 'CRLF Ln 2, Col 46'.

Task 13. Update airline ID 5

I changed airline 5 to Global Airways, set the country to United Kingdom, and updated updated_at to the current timestamp.



The screenshot displays the pgAdmin 4 interface. On the left, the Object Explorer shows a tree structure with 'Servers', 'lab2', and 'phonebook'. The main pane is titled 'airport_db/postgres@lab2' and contains a SQL query editor. The query is as follows:

```
1 UPDATE airline
2 SET airline_name = 'Global Airways',
3     airline_country = 'United Kingdom',
4     updated_at = CURRENT_TIMESTAMP
5 WHERE airline_id = 5;
```

Below the query editor, the 'Messages' tab is active, showing the execution result:

```
UPDATE 1
Query returned successfully in 35 msec.
```

The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.035' and 'CRLF Ln 5, Col 22'.

Task 14. Set Capital District

I set state to Capital District for airports in Astana, London and Tokyo.

The screenshot shows the pgAdmin 4 web interface. On the left, the 'Object Explorer' pane displays a tree structure with 'Servers' expanded, showing 'lab2' and 'phonebook'. The main workspace is titled 'airport_db/postgres@lab2'. It contains a 'Query' editor with the following SQL code:

```
1 UPDATE airport
2 SET state = 'Capital District'
3 WHERE city IN ('Astana', 'London', 'Tokyo');
```

Below the query editor, the 'Messages' tab is active, showing the execution result:

```
UPDATE 1
Query returned successfully in 36 msec.
```

A green notification box at the bottom right of the interface states: '✓ Query returned successfully in 36 msec. ✕'. The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.036' and 'CRLF Ln 3, Col 45'.

Task 15. Mark March 2024 checks as Checked

I changed baggage checks created in March 2024 from Not checked to Checked and updated the timestamp.

The screenshot displays the pgAdmin 4 web interface. The top navigation bar includes 'File', 'Object', 'Tools', 'Edit', 'View', 'Window', and 'Help'. The left sidebar shows the 'Object Explorer' with a tree view containing 'Servers', 'lab2', and 'phonebook'. The main workspace is titled 'airport_db/postgres@lab2' and contains a SQL query editor with the following code:

```
1 UPDATE baggage_check
2 SET check_result = 'Checked',
3   updated_at = CURRENT_TIMESTAMP
4 WHERE created_at >= '2024-03-01'
5    AND created_at < '2024-04-01'
6    AND check_result = 'Not checked';
```

Below the query editor, the 'Messages' tab is active, showing the execution result: 'UPDATE 0' and 'Query returned successfully in 40 msec.' A green notification box at the bottom right confirms: '✓ Query returned successfully in 40 msec. ✕'. The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.040' and 'CRLF Ln 6, Col 36'.